WORKSHEET 8.8

Solids of Revolution: Discs

Unit 8: Applications of Integration

Key ideas

- When a region that touches the axis is revolved, each slice is a disc: $V = \pi \int_a^b R^2\,dx$.
- R is the distance from the curve to the axis of revolution.
- About a horizontal line $y = k$: $R = |f(x) - k|$, and integrate in x.
- About the y-axis or a vertical line: write the curve as $x = \ldots$ and integrate in y.

Worked examples

Example 1 · About the x-axis

The region under $y = \dfrac{2}{x}$ for $1 \le x \le 4$ is revolved about the x-axis. Find the volume.

Solution. $\pi \displaystyle\int_1^4 \frac{4}{x^2}\,dx = 4\pi\left(1 - \frac{1}{4}\right)$.

3π.

Example 2 · About the y-axis

The region in the first quadrant bounded by $y = x^4$, the y-axis and $y = 1$ is revolved about the y-axis. Find the volume.

Solution. $R = x = y^{1/4}$: $\pi \displaystyle\int_0^1 y^{1/2}\,dy$.

$\dfrac{2\pi}{3}$.

Example 3 · About another line

The region enclosed by $y = x^2$ and $y = 4$ is revolved about the line $y = 4$. Find the volume.

Solution. $R = 4 - x^2$: $\pi \displaystyle\int_{-2}^{2} (4 - x^2)^2\,dx$.

$\dfrac{512\pi}{15}$.

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Practice

CORE

1. The region under $y = 3$ for $0 \leq x \leq 5$ is revolved about the x-axis. Find the volume.

2. The region under $y = x$ for $0 \leq x \leq 3$ is revolved about the x-axis. Find the volume.

3. The region under $y = \sqrt{x+1}$ for $0 \leq x \leq 3$ is revolved about the x-axis. Find the volume.

4. Write, but do not evaluate, an integral for the volume when the region under $y = \cos x$ for $0 \leq x \leq \dfrac{\pi}{2}$ is revolved about the x-axis.

EXAM LEVEL

5. The region under $y = \sec x$ for $0 \leq x \leq \dfrac{\pi}{4}$ is revolved about the x-axis. Find the volume.

6. The region under $y = 2\sqrt{\cos x}$ for $0 \leq x \leq \dfrac{\pi}{2}$ is revolved about the x-axis. Find the volume.

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7. The region bounded by $y = \ln x$, the y-axis, $y = 0$ and $y = 2$ is revolved about the y-axis. Find the volume.

8. The region bounded by $y = \sqrt{x}$, the x-axis and $x = 4$ is revolved about the line $x = 4$. Find the volume.

Question 9 · Free-response practice

Let R be the region in the first quadrant bounded by $y = \sqrt{9 - x}$ and the coordinate axes.

(a) Find the area of R.

(b) Find the volume of the solid generated when R is revolved about the x-axis.

(c) Find the volume of the solid generated when R is revolved about the y-axis.

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Worked solutions

1. A cylinder: $\pi(3)^2(5) = 45\pi$.

2. $\pi \displaystyle\int_0^3 x^2\,dx = 9\pi$.

3. $\pi \displaystyle\int_0^3 (x+1)\,dx = \frac{15\pi}{2}$.

4. $\pi \displaystyle\int_0^{\pi/2} \cos^2 x\,dx$.

5. $\pi\Big[\tan x\Big]_0^{\pi/4} = \pi$.

6. $\pi \displaystyle\int_0^{\pi/2} 4\cos x\,dx = 4\pi$.

7. $x = e^y$: $\pi \displaystyle\int_0^2 e^{2y}\,dy = \frac{\pi(e^4-1)}{2}$.

8. $R = 4 - y^2$ for $0 \le y \le 2$: $\pi \displaystyle\int_0^2 (16 - 8y^2 + y^4)\,dy = \pi\left(32 - \frac{64}{3} + \frac{32}{5}\right) = \frac{256\pi}{15}$.

9. (a) $\displaystyle\int_0^9 \sqrt{9-x}\,dx = \frac{2}{3}(27) = 18$.

   (b) $\pi \displaystyle\int_0^9 (9-x)\,dx = \frac{81\pi}{2}$.

   (c) $x = 9 - y^2$ for $0 \le y \le 3$: $\pi \displaystyle\int_0^3 (9-y^2)^2\,dy = \pi\left(243 - 162 + \frac{243}{5}\right) = \frac{648\pi}{5}$.

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